

Lecture - 11

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

TEST SIGNIFICANCE OF SAMPLE MEAN

we have already pointed out that one basic difference between Z-test and t-test is that Z-test is used when population SD is known whether sample size is large or small and t-test is used when population SD is unknown whether sample size is small or large. But in case of large sample size Z-test is an appropriate of t-test as we did in previous unit. But in practice standard deviation of population is not known and sample size is small so in this situation, we use t-test provided population under study is normal.

Assumptions

Virtually every test has some assumptions which must be met prior to the application of the test. This t-test needs following assumptions to work:

- i. The characteristic under study follows normal distribution. In other words, populations from which random sample is drawn should be normal with respect to the characteristic of interest.
- ii. Sample observations are random and independent.

iii. Population variance σ^2 is unknown.

For describing this test, let $X_1, X_2, \dots, X_n$ be a random sample of small size n (< 30) selected from a normal population with mean μ and unknown variance σ^2 .

Now, follow the same procedure as we have discussed in previous section, that is, first of all we setup the null and alternative hypotheses. Here, we want to test the claim about the specified value μ_0 of population mean μ so we can take the null and alternative hypotheses as

$$H_0: \mu = \mu_0 \text{ and } H_1: \mu \neq \mu_0 \quad [\text{for two - tailed test}]$$

[Here $\theta = \mu$ and $\theta_0 = \mu_0$ if we compare it with general procedure]

$$\left. \begin{array}{l} H_0: \mu \leq \mu_0 \text{ and } H_1: \mu \geq \mu_0 \\ H_0: \mu \geq \mu_0 \text{ and } H_1: \mu \leq \mu_0 \end{array} \right\} [\text{for one - tailed test}]$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}} \sim t_{n-1} \quad \text{under } H_0$$

Where, $\bar{X} = \frac{1}{n} \sum X_i$ is the sample mean and $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is the sample variance.

For computational simplicity, we may use the following formulae for $\bar{X}, S^2$:

$$\bar{X} = a + \frac{1}{n} \sum d \text{ and } S^2 = \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right]$$

where, $d = (X - a)$, 'a' being the assumed arbitrary value.

Here, the test statistic t follows t -distribution with $(n - 1)$ degrees of freedom.

After substituting values of $\bar{X}$, S and n , we get calculated value of test statistic t . Then we look for critical (or cut-off or tabulated) value(s) of test statistic t from the t -table. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed in previous section.

Let us do some examples of testing of hypothesis about population mean using t -test.

Example 1: A manufacturer claims that a special type of projector bulb has an average life 160 hours. To check this claim an investigator takes a sample of 20 such bulbs, puts on the test, and obtains an average life 167 hours with standard deviation 16 hours. Assuming that the life time of such bulbs follows normal distribution, does the investigator accept the manufacturer's claim at 5% level of significance?

Solution: Here, we are given that $\mu_0 = 160$, $n = 20$, $\bar{X} = 167$ and $S = 16$. Here, we want to test the manufacturer claims that a special type of projector bulb has an average life (μ) 160 hours. So claim is $\mu = 160$ and its complement is $\mu \neq 160$.

Since the claim contains the equality sign so we can take the claim as the null hypothesis and complement as the alternative hypothesis. Thus,

$$H_0: \mu = \mu_0 = 160 \text{ and } H_1: \mu \neq 160$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Here, we want to test the hypothesis regarding population mean when population SD is unknown. Also sample size is small $n = 20$ ($n < 30$) and population under study is normal, so we can go for t-test for testing the hypothesis about population mean.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}} = \frac{167 - 160}{16/\sqrt{20}} = \frac{7}{3.58} = 1.96$$

The critical value of the test statistic t for various df and different level of significance α are given in Table.

The critical (tabulated) values of test statistic for two-tailed test corresponding $(n-1) = 19$ df at 5% level of significance are $\pm t_{n-1, \frac{\alpha}{2}} = \pm t_{19, 0.025} = \pm 2.093$.

Since calculated value of test statistic t (= 1.96) is greater than the critical value (= - 2.093) and is less than critical value (= 2.093), that means calculated value

of test statistic lies in non-rejection region. So we do not reject the null hypothesis i.e. we support the manufacture's claim at 5% level of significance.

Decision according to p-value

Since calculated value of test statistic t is based on 19 df therefore, we use row for 19 df in the t -table and move across this row to find the values in which calculated t -value falls. Since calculated t -value falls between 1.729 and 2.093 corresponding to one-tailed area $\alpha = 0.05$ and 0.025 respectively therefore p -value lies between 0.025 and 0.05, that is,

$$0.025 < p\text{-value} < 0.05$$

Since test is two-tailed so

$$2 \times 0.025 = 0.05 < p\text{-value} < 0.10 = 0.05 \times 2$$

Since p -value is greater than $\alpha = (0.05)$ so we do not reject the null hypothesis at 5% level of significance.

Thus, we conclude that sample fails to provide us sufficient evidence against the null hypothesis so we may assume that the manufacture's claim is true so the investigator may accept the manufacturer's claim at 5% level of significance.

Now, you can try the following exercises.

Example 2: The mean share price of companies of Pharma sector is Rs.70. The share prices of all companies were changed time to time. After a month, a sample of 10 Pharma companies was taken and their share prices were noted as below:

70, 76, 75, 69, 70, 72, 68, 65, 75, 72

Assuming that the distribution of share prices follows normal distribution, test whether mean share price is still the same at 1% level of significance?

Example 3: A tyre manufacturer claims that the average life of a particular category of his tyre is 18000 km when used under normal driving conditions. A random sample of 16 tyres was tested. The mean and SD of life of the tyres in the sample were 20000 km and 6000 km respectively. Assuming that the life of the tyres is normally distributed, test the claim of the manufacturer at 1% level of significance using appropriate test.

Example 4: It is known that the average weight of cadets of a centre follows normal distribution. Weights of 10 randomly selected cadets from the same centre are as given below:

48, 50, 62, 75, 80, 60, 70, 56, 52, 77

Can we say that average weight of all cadets of the centre from which the above sample was taken is equal to 60 kg at 5% level of significance?

Chandigarh University